

Charan Tej Matam

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EDUCATION

IITDM Kurnool

Bachelor of Technology in Computer Science and Engineering

August 2024 – Present

CGPA: 8.55/10

Sr Jr. College

Intermediate in M.P.C

April 2022 – April 2024

Kurnool, AP | Marks: 975/1000

PROJECTS

AI Notes Enhancer

February 2026

Technologies: Python, Natural Language Processing (NLP), Generative AI APIs, Streamlit, Git

- Developed a full-stack AI-powered application that ingests, enhances, and summarizes raw educational study notes using Large Language Models (LLMs).
- Implemented rigorous text preprocessing tokenization and advanced prompt engineering techniques to output structured markdown notes.
- Integrated generative AI orchestration workflows to support automatic core concept refinement, flashcard generation, and data organization.
- Designed and deployed an interactive, low-latency web interface using Streamlit to support real-time PDF and text uploads.
- Enhanced student readability statistics and overall learning efficiency by compacting verbose textbook chapters into high-yield summaries.

Compiler Design Engine

November 2025

Technologies: C++, Compiler Architecture, Lexical Analysis, Context-Free Grammars (CFGs)

- Developed core architecture modules of a customized programming language compiler, including structural phases from lexing to parsing.
- Implemented optimized tokenization algorithms leveraging Deterministic Finite Automata (DFA) to classify keywords, operators, and literals.
- Designed recursive-descent parsing modules to syntactically validate program source streams against defined structural grammar rules.
- Created an integrated compile-time error reporting subsystem capable of identifying and outputting localized syntax violation details.
- Applied theoretical computer science principles covering LL(1) parsing tables, symbol tree tables, and abstract syntax trees (ASTs).

TECHNICAL SKILLS

- **Languages:** C++, Python, JavaScript, MySQL, HTML5, CSS3
- **Machine Learning & AI:** Linear Regression, Logistic Regression, K-Means Clustering, Tokenization, LLM APIs
- **Data Structures & Algorithms:** Arrays, Linked Lists, Stacks, Queues, Trees, Binary Search, Sorting, Graphs
- **Backend & Web Development:** Node.js, Express.js, FastAPI, RESTful APIs, Streamlit Interface Architecture
- **Libraries & Frameworks:** Pandas, NumPy, Matplotlib, LangChain Orchestration, Keras, TensorFlow
- **Tools & Platforms:** Git, GitHub Version Control, Docker, Google Colab, Visual Studio Code, Postman API Framework

SOFT SKILLS

- **Communication:** Proven ability to clarify complex engineering concepts and present architectures comprehensively to technical groups.
- **Team Collaboration:** Collaborated effectively with multi-disciplinary engineering peers during project cycles and technical hackathons.
- **Adaptability & Grit:** Highly self-motivated loop learner eager to pick up emerging engineering technologies like RAG frameworks and Vector Datastores.